

Traffic Signal Violation Detection at Simpang Wirosaban Using YOLOv11, HSV Thresholding, and SORT

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Abstract—This is a placeholder abstract. Replace this text with a concise summary of the research background, method, main findings, and conclusion.

I. INTRODUCTION

A. Background

Traffic signal violations, particularly red light running, constitute one of the most prevalent and dangerous forms of traffic misconduct at urban intersections. [1] In Indonesia, the problem has reached a critical scale. Data from the National Criminal Information Center of the Indonesian National Police recorded 148,798 traffic accidents throughout 2023, of which 94.63% were attributed to human error [2]. The Director of Traffic Accident Investigation of the Indonesian National Police further stated that every accident is preceded by a violation, underscoring the direct causal relationship between traffic infractions and collision events [2]. Simpang Wirosaban, a signalized intersection in Yogyakarta, represents a typical urban intersection in Indonesia where such violations occur at high frequency, particularly during nighttime hours when traffic enforcement is minimal. The absence of automated monitoring at such locations necessitates a technology-driven approach to continuous, objective violation detection.

Conventional methods of traffic enforcement rely on human officers stationed at intersections or on radar-based sensors embedded in road infrastructure. These approaches are costly to deploy and maintain, limited in spatial coverage, and susceptible to human fatigue and inconsistency [3]. The proliferation of closed-circuit television (CCTV) infrastructure at intersections across Indonesian cities presents an opportunity to repurpose existing hardware for automated enforcement

without additional installation costs. Computer vision techniques applied to CCTV footage have demonstrated considerable promise in this domain. Ong et al. [3] proposed a YOLO-based system targeting red light crossing and double-line overtaking violations from CCTV footage, establishing a foundational framework for vision-based violation detection. Similarly, Tomas et al. [4] demonstrated that YOLOv5 combined with Faster R-CNN achieved high accuracy in detecting red light and wrong-way violations at intersections, reinforcing the viability of deep learning pipelines for this task.

Vehicle detection in video-based systems has been significantly advanced by the YOLO family of single-stage object detectors. Since its introduction, YOLO has undergone continuous architectural refinement, with each iteration improving detection accuracy and inference speed. YOLO11, released by Ultralytics in September 2024 at YOLO Vision 2024, introduced C3K2 blocks, an enhanced Spatial Pyramid Pooling Fast (SPPF) module, and a C2PSA partial spatial attention mechanism [5]. The YOLO11m variant achieves higher mean average precision on the COCO benchmark with 22% fewer parameters than its YOLOv8m predecessor, making it well-suited for real-time deployment on constrained hardware [5]. For multi-object tracking across video frames, this study employs the Simple Online and Realtime Tracking (SORT) algorithm proposed by Bewley et al. [6], which combines the Kalman filter for state estimation with the Hungarian algorithm for frame-to-frame data association. Despite its computational simplicity, SORT achieves tracking performance comparable to significantly more complex methods while operating at over 260 Hz, satisfying the latency requirements of real-time traffic monitoring.

A persistent challenge in nighttime traffic surveillance is the degradation of image quality caused by uneven illumination, vehicle headlight glare, and sensor noise, all of which reduce object detection confidence. Lashkov et al. [7] demonstrated that Contrast Limited Adaptive Histogram Equalization (CLAHE) applied as a preprocessing step substantially improves vehicle detection performance in nighttime conditions. Unlike standard histogram equalization, CLAHE operates on localized image tiles and enforces a contrast limit to prevent noise amplification, making it particularly effective for scenes with spatially non-uniform lighting such as signalized intersections at night [8], [9]. In this study, CLAHE is applied exclusively to the luminance channel of the LAB color space prior to YOLO11 inference, preserving the chrominance information required by the HSV-based traffic light detector. This decoupled enhancement strategy ensures that preprocessing improvements to vehicle detection do not introduce artifacts into the color-sensitive components of the pipeline.

B. Objective

The primary objectives of this study include:

- To develop a real-time traffic violation detection system capable of identifying vehicles that cross the stop line during a red light signal using CCTV footage.
- To implement vehicle detection and multi-object tracking using YOLOv11 and SORT algorithm to accurately localize and track vehicles within a defined detection zone.
- To design a rule-based violation logic that determines infractions based on vehicle trajectory relative to the stop line and the current traffic light state.
- To evaluate the effectiveness of image enhancement techniques, including CLAHE, Gamma Correction.

C. Scope

The scope of the study includes:

- The study only can be implemented at the Simpang Wirosaban, Yogyakarta, using footage from a fixed CCTV camera.
- The system is optimized for nighttime conditions, where traffic light colors are more distinguishable.
- Detection performance may decrease under rainy weather, poor visibility, or varying lighting conditions.

II. LITERATURE REVIEW

A. Related Work

B. HSV (Hue, Saturation, Value) Color Model

In the field of computer vision, selecting an appropriate color space is critical for the accuracy of object detection under varying environmental conditions. The HSV (Hue, Saturation, Value) color model is a cylindrical representation of colors that more closely aligns with human visual perception compared to the standard RGB model. While RGB defines colors as a combination of primary light sources, it is highly sensitive to changes in brightness, shadows, and external lighting. The

HSV model decouples color information from intensity, using three distinct components:

- Hue (H): Represents the dominant wavelength or the actual "color" (e.g., Red, Yellow, Green) expressed in degrees from 0° to 360° .
- Saturation (S): Indicates the vibrancy or purity of the color, ranging from 0 to 100%.
- Value (V): Represents the brightness or luminance of the color.

C. You Only Look Once 11 (YOLO11)

YOLO11 is a single-stage object detection model released by Ultralytics in September 2024 at the YOLO Vision 2024 event [5]. It follows the conventional backbone-neck-head architecture common to the YOLO family, with three principal architectural innovations that distinguish it from its predecessors.

The backbone employs C3k2 blocks (Cross Stage Partial with kernel size 2) as a replacement for the C2f blocks used in YOLOv8. C3k2 implements the CSP bottleneck using two convolutions with smaller kernels, reducing parameter count while preserving gradient flow across stages [10]. Multi-scale feature aggregation is handled by the Spatial Pyramid Pooling Fast (SPPF) module in the neck, which applies sequential max-pooling operations of fixed kernel size and concatenates the results, enabling the model to capture context at multiple receptive field scales without the computational overhead of the original SPP formulation. Following SPPF, the C2PSA module (Convolutional block with Parallel Spatial Attention) incorporates a self-attention mechanism that allows the model to selectively weight spatial regions of the feature map, improving sensitivity to small and partially occluded objects [10]. The detection head retains the decoupled structure introduced in YOLOv8, maintaining separate branches for classification and bounding box regression.

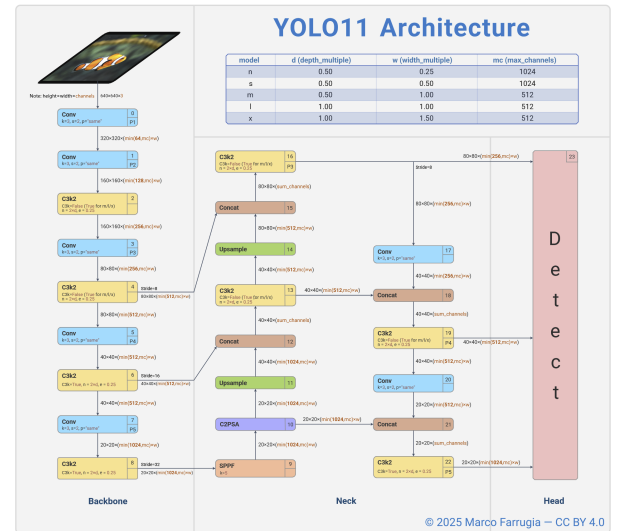


Fig. 1: The architecture of YOLO11 [11].

Compared to YOLOv8m, YOLO11m achieves higher mean Average Precision (mAP) on the COCO benchmark while using 22% fewer parameters [5], making it well suited for real-time deployment on GPU hardware. In this study, the medium variant (YOLO11m) is used with inference restricted to five COCO vehicle classes: bicycle, car, motorcycle, bus, and truck.

D. Simple Online and Realtime Tracking (SORT)

SORT is a multi-object tracking algorithm proposed by Bewley et al. [6] that follows the tracking-by-detection paradigm, where an external detector supplies bounding boxes each frame and the tracker is responsible solely for associating detections across time. Despite relying exclusively on bounding box geometry without any appearance features, SORT achieves tracking accuracy comparable to far more complex methods while operating at over 260 Hz [6].

Each tracked object is represented by a `KalmanBoxTracker` with a 7-dimensional state vector:

$$\mathbf{x} = [c_x, c_y, s, r, \dot{c}_x, \dot{c}_y, \dot{s}]^T \quad (1)$$

where c_x and c_y are the centroid coordinates of the bounding box, s is the box area, r is the aspect ratio, and $\dot{c}_x$, $\dot{c}_y$, $\dot{s}$ are their respective first-order velocities. The aspect ratio r is assumed constant. The Kalman filter employs a constant velocity linear motion model as the state transition, with the observation vector consisting of $[c_x, c_y, s, r]^T$ only. At each frame, every active tracker advances its state estimate through the Kalman prediction step before any new detections are considered.

Association between predicted tracker states and incoming detections is formulated as a linear assignment problem. An IoU cost matrix is constructed between all predicted bounding boxes and all current detections, and the optimal one-to-one assignment is solved using the Hungarian algorithm [6]. A detection-tracker pair is accepted only if their IoU exceeds a minimum threshold; pairs below this threshold are treated as unmatched. Matched trackers update their Kalman states with the observed measurement. Unmatched detections initialize new trackers, and trackers that remain unmatched beyond a maximum age threshold are removed. A tracker is only reported as a confirmed track once it has been matched for a minimum number of consecutive frames, suppressing spurious short-lived detections.

III. METHODOLOGY

A. Proposed Approach

This study proposes a computer vision-based approach for detecting traffic signal violations using CCTV footage from a static camera installed at Simpang Wirosaban, Yogyakarta. The system is designed as a multi-stage pipeline (Figure 2) consisting of image enhancement, vehicle detection, multi-object tracking, traffic light recognition, and violation analysis. To reduce computational overhead and eliminate detections from irrelevant regions such as sidewalks and surrounding

infrastructure, a polygon-based Region of Interest (ROI) is defined to constrain the analysis strictly to the active roadway area.

Prior to vehicle detection, each frame undergoes a three-stage image enhancement pipeline to improve visibility under nighttime and uneven lighting conditions. Gamma correction is first applied to brighten underexposed regions by adjusting pixel intensity according to a tunable gamma value ($\gamma = 1.5$). The gamma-corrected frame is then passed to Contrast Limited Adaptive Histogram Equalization (CLAHE), applied exclusively to the luminance channel of the LAB color space to enhance local contrast without introducing chrominance distortion. Finally, Unsharp Masking is applied to sharpen object boundaries by subtracting a Gaussian-blurred version of the frame from the original, which mitigates the effect of motion blur on moving vehicles. This enhancement pipeline is applied solely to the input of the vehicle detector; the original unmodified frame is retained for all other processing stages to preserve color fidelity.

Within the defined ROI, vehicle detection is performed using YOLO11, which identifies vehicles present inside the polygonal area on each frame. The detected bounding boxes are subsequently processed by the Simple Online and Realtime Tracking (SORT) algorithm, which employs a Kalman filter for state prediction and the Hungarian algorithm for frame-to-frame identity association. This tracking mechanism assigns a persistent unique identifier to each vehicle and maintains its trajectory across consecutive frames, enabling reliable monitoring of individual vehicle movements over time.

In parallel with vehicle detection, the traffic light state is determined using an HSV color-based thresholding method applied to a dedicated ROI positioned at the traffic light location in the frame. The detected state is used as a conditional input to the violation logic module. A virtual stop line is defined within the ROI to represent the legal stopping boundary at the intersection. Violation detection is triggered when a tracked vehicle, whose trajectory originated from within the ROI, crosses the stop line at the moment the traffic light is in a red state. To prevent duplicate records, each unique vehicle identity is registered as a violation at most once. The system produces output in the form of annotated video frames displaying bounding boxes, vehicle identities, stop line geometry, and a running violation counter, providing a structured and automated means of traffic violation documentation.

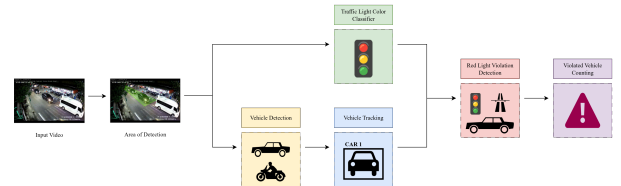


Fig. 2: Diagram flow of traffic signal violation detection at Simpang Wirosaban.

B. Data Acquisition

The video data used in this study was obtained from a publicly accessible CCTV stream provided by the Yogyakarta Provincial Government through the official traffic monitoring portal at <https://cctv.jogjapro.go.id>, specifically from the Simpang Wirosaban (RSUD) Timur View camera. The footage was recorded at a resolution of 1920×1200 pixels under two distinct lighting conditions, namely daytime and nighttime, to ensure that the system is evaluated against varying illumination environments representative of real-world deployment scenarios.

The acquired footage serves two primary purposes in this study. First, a subset of frames extracted from the video is used to manually define the polygon-based Region of Interest (ROI) that delimits the active roadway area subject to monitoring. The ROI polygon is specified by selecting boundary coordinates that enclose the relevant lane area while excluding sidewalks, parked vehicles, and surrounding infrastructure. The virtual stop line is also derived from this process, defined as a segment spanning two vertices of the ROI polygon that corresponds to the legal stopping boundary at the intersection. Second, the remaining footage is used directly for system inference, serving as the primary input to the full detection and violation analysis pipeline during both development and evaluation phases.

C. Image Enhancement

To improve vehicle detection performance under nighttime and uneven lighting conditions, a three-stage image enhancement pipeline is applied to each frame prior to YOLO11 inference. The pipeline consists of Gamma Correction, Contrast Limited Adaptive Histogram Equalization (CLAHE), and Unsharp Masking, applied sequentially. The original unmodified frame is retained separately for traffic light detection and output rendering.

1) *Gamma Correction*: Gamma correction adjusts the overall brightness of a frame through a nonlinear pixel intensity transformation defined in Equation 2:

$$I_{out} = 255 \cdot \left(\frac{I_{in}}{255} \right)^{1/\gamma} \quad (2)$$

where I_{in} is the original pixel intensity and $\gamma = 1.5$. A value of $\gamma > 1$ brightens the image by elevating mid-tone and shadow regions, which improves the visibility of vehicles in underexposed areas of the frame. The transformation is implemented as a precomputed lookup table (LUT) for computational efficiency.

2) *CLAHE*: Following gamma correction, CLAHE is applied to enhance local contrast. Unlike global histogram equalization, CLAHE divides the image into a grid of non-overlapping tiles and performs histogram equalization independently within each tile, with a contrast limit (`clipLimit` = 2.0) that prevents noise amplification in near-uniform regions [8]. To avoid introducing color distortion, CLAHE is applied exclusively to the luminance channel L of the LAB

color space, while the chrominance channels A and B remain unmodified. The enhanced L channel is then merged back before converting the frame to BGR.

3) *Unsharp Masking*: Unsharp masking is applied as the final enhancement step to sharpen object boundaries degraded by motion blur. The operation subtracts a Gaussian-blurred version of the frame from the original according to Equation 3:

$$I_{sharp} = (1 + \alpha) \cdot I - \alpha \cdot G_{\sigma}(I) \quad (3)$$

where G_{σ} denotes Gaussian blurring with $\sigma = 1.0$ and $\alpha = 1.5$ controls sharpening strength. The result accentuates high-frequency edge information, improving the definition of vehicle contours in motion-blurred nighttime frames.

D. Vehicle Detection Using YOLO11

Vehicle detection is performed using YOLO11m, a medium-sized variant of the YOLO11 architecture provided by Ultralytics. The model is loaded via the `ultralytics` library and configured with a confidence threshold of 0.35 and an IoU threshold of 0.4. Detection is restricted to five COCO class indices corresponding to vehicle categories: bicycle (1), car (2), motorcycle (3), bus (5), and truck (7).

For each frame, the model performs inference on the enhanced frame and returns bounding boxes in `xyxy` format along with their confidence scores. Detections whose centroid falls outside the detection polygon are discarded using `cv2.pointPolygonTest` prior to being passed to the tracker. A secondary Non-Maximum Suppression (NMS) step is applied post-inference with an IoU threshold of 0.5 to suppress redundant overlapping boxes that may arise from adjacent detections of the same vehicle.

E. Vehicle Tracking Using SORT

At each frame, every active tracker predicts its next state via the Kalman filter's linear motion model. The predicted bounding boxes are then matched against YOLO11 detections using the Hungarian algorithm applied to a pairwise IoU cost matrix. A detection-tracker pair is accepted if their IoU exceeds 0.25 (`iou_threshold`); otherwise both are treated as unmatched. Unmatched detections spawn new trackers, while matched trackers update their Kalman state with the observed measurement.

A tracker is only included in the output if it has been matched for at least `min_hits` = 2 consecutive frames, which suppresses short-lived false detections. A tracker is deleted if it has not been matched for more than `max_age` = 30 frames. Each confirmed tracker is assigned a globally unique integer identity that persists across frames, forming the trajectory used by the violation logic module.

F. Traffic Light Color Classifier Using HSV thresholding

Write the detection and tracking method here.

G. Traffic Signal Violation Logic

A violation is recorded when a tracked vehicle crosses the virtual stop line while the traffic light state is RED (Figure 3). The stop line is defined as a segment connecting two vertices of the detection polygon, specifically from coordinate (916, 762) to (1260, 586).

For each tracked vehicle, the system computes the signed side of the stop line using the cross product formulation in Equation 4:

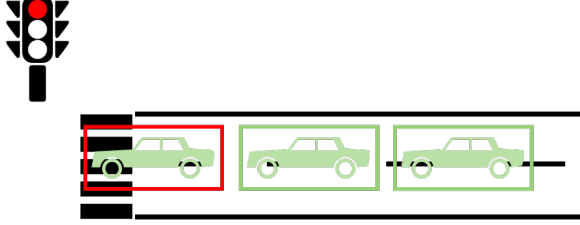


Fig. 3: Violation detection illustration.

$$d = (x_2 - x_1)(c_y - y_1) - (y_2 - y_1)(c_x - x_1) \quad (4)$$

where (x_1, y_1) and (x_2, y_2) are the endpoints of the stop line, and (c_x, c_y) is the vehicle centroid. The sign of d determines which side of the line the centroid lies on. A crossing event is detected when the sign of d changes between two consecutive frames, indicating that the vehicle has traversed the stop line.

To suppress false positives arising from tracking noise or detections outside the roadway, two additional validity conditions are enforced. First, the midpoint between the vehicle's previous and current positions must be within 150 pixels of the stop line segment. Second, at least one of the two positions must lie within the detection polygon. A crossing that satisfies both conditions is registered to `crossed_ids`. If the traffic light state at the moment of crossing is RED, the vehicle identity is added to `violation_ids` and the violation counter is incremented.

To handle tracker ID reassignment caused by occlusion or missed detections, a ghost tracking mechanism is implemented. When a raw tracker ID disappears from the active track list, its last known position is retained for up to 60 frames (`ghost_ttl = 60`). If a new raw ID appears within 60 pixels (`remap_threshold = 60`) of a ghost's last position, it is remapped to the original canonical ID, preserving trajectory continuity across ID switches.

H. Violated Vehicle Counting

Each vehicle identity is permitted to register at most one violation regardless of how many times it crosses the stop line during a session. This constraint is enforced by checking membership in `violation_ids` prior to incrementing the counter. The total violation count is maintained as a running integer (`violation_count`) and is rendered directly onto

each output frame alongside the current traffic light state and timestamp.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The experiments were conducted using input videos with a resolution of 1920×1200 pixels to preserve fine-grained details of vehicle motion and traffic light status. All video processing and model inference were executed on a workstation equipped with an NVIDIA Tesla T4 GPU, which provided hardware acceleration for YOLO11 detection and SORT-based tracking during the evaluation stage.

B. Results

Write the main results here.

V. CONCLUSION

A. Conclusion

Write the final conclusion here.

B. Future Work

Write possible future improvements here.

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